is approximately 0.3 V for germanium and 0.7 V for silicon. The current flow is negligible when the applied voltage is less than the threshold voltage. Beyond the threshold voltage, increase in current is significant even for a small increase in voltage.

(ii) The graph clearly infers that the current flow is not linear and is exponential. Hence it does not obey Ohm's law.

(iii) The forward resistance (r_F) of the diode is the ratio of the small change in voltage (ΔV_F) to the small change in current (ΔI_F). That is, $r_F = \frac{\Delta V_F}{\Delta I_F}$.

(iv) Thus the diode behaves as a conductor when it is forward biased.

However, if the applied voltage is increased beyond a rated value, it will produce an extremely large current which may destroy the junction due to overheating. This is called as the breakdown of the diode and the voltage at which the diode breaks down is called the breakdown voltage. Thus, it is safe to operate a diode between the threshold voltage and the breakdown voltage.

ii) Reverse characteristics

The circuit to study the reverse characteristics is shown in Figure 10.15(a). In the reverse bias, the *p*-side of the diode is connected to the negative terminal and *n*-side to the positive terminal of the dc power supply.

A graph drawn between the reverse bias voltage and the current across the junction is called the reverse characteristics of a *p-n* junction diode. It is shown in Figure 10.15(b). Under this bias, a very small

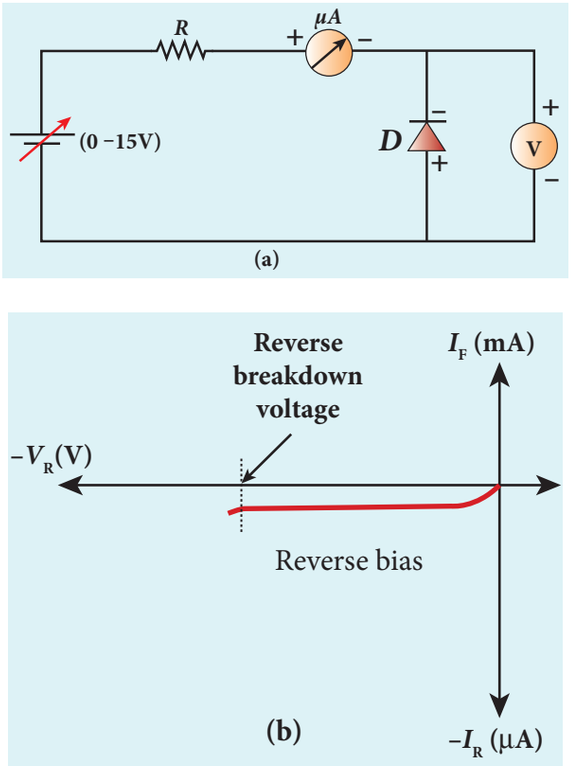


Figure 10.15 *p-n* junction diode
(a) Diode under reverse bias
(b) Reverse characteristics

current in μA flows across the junction. This is due to the flow of the minority charge carriers and is called the leakage current or reverse saturation current. This reverse current is independent of the voltage up to a certain voltage, known as breakdown voltage.

Note

Ideal diode: It acts like a conductor when it is forward biased. When it is reverse biased, it acts like an insulator. For ideal diodes, the forward resistance is zero and barrier potential is considered negligible.

The forward and reverse charactristics are given in one graph as shown in Figure 10.16.

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